- Equation of Straight line

  y = mx + c

  ax +by + c =0

  $W^T\ x + c = 0$

- Use of W.V (to know if the data point is pn +ve or –ve side).
- Sigmoid Activation Fuction (to shrink the data form 0 to 1)
- Etropy Gini Impurity

Components required to create strait line

- x and y point
- Slope
- Intercept

Components required to create strait line

- x =1
- y =1
- Slope = 1.5

$y = mx + c$

Slope

$$m = \frac{\text{Change in Y}}{\text{Change in X}}$$

$$\text{Slope } (m) = \frac{Y_2 - Y_1}{X_2 - X_1}$$

X	Y
2	10
4	20

Y1 = 10

Y2= 20

X1 = 2

X2 =4

$$m = \frac{20 - 10}{4 - 2}$$

$$m = \frac{10}{2}$$

$$\boxed{m = 5}$$

Equation of Straight Line

$y = m x + c$

m (Slope or Coefficient) = slope of the line

c Intercept (where the best-fit line meets the Y axis).

M=1.3

C=0

x	m x + c = y
1.0	(1.3 * 1) + 0 = 1.3
3.0	(1.3 * 3) + 0 = 3.9
4.0	(1.3 * 4) + 0 = 5.2
6.0	(1.3 * 6) + 0 = 7.8
7.0	(1.3 * 7) + 0 = 9.1
8.0	(1.3 * 8) + 0 = 10.4

M=1.2

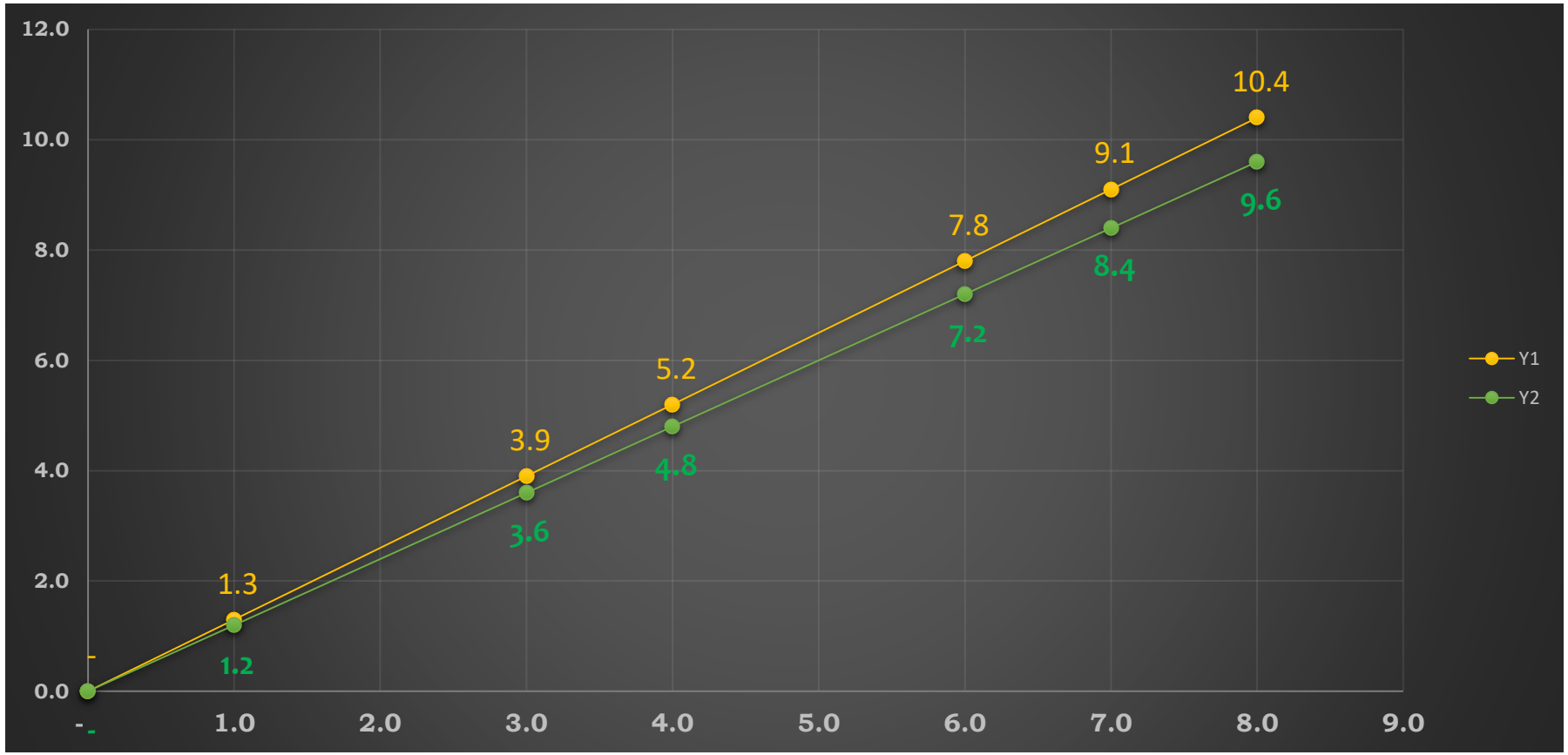
C=0

m x + c = y
(1.2 * 1) + 0 = 01.2
(1.2 * 3) + 0 = 3.6
(1.2 * 4) + 0 = 4.8
(1.2 * 6) + 0 = 7.2
(1.2 * 7) + 0 = 8.4
(1.2 * 8) + 0 =9.6

M=1.3
C=0

x	y _1	Y_2
1.0	1.3	01.2
3.0	3.9	3.6
4.0	5.2	4.8
6.0	7.8	7.2
7.0	9.1	8.4
8.0	10.4	9.6

M=1.2
C=0



Equation of Straight Line

$y = \mathbf{m}\,x + \mathbf{c}$

$y = \mathbf{m}\,x + \mathbf{c}$

Equation of Straight Line

$$ax + by + c = 0$$

$$by + c = -ax$$

$$by = -ax + (-c)$$

$$y = \frac{-a}{b}x + \frac{-c}{b}$$

m OR $\frac{-a}{b}$ = (Slope or Coefficient) slope of the line

c OR $\frac{-c}{b}$ = Intercept (where the best-fit line meets the Y axis).

$$-1x + 2y - 2 = 0$$

$$-1x + 2y - 2 = 1$$

$$-1x + 2y - 3 = 0$$

$$-1x + 2y - 2 = -1$$

$$-1x + 2y - 1 = 0$$

$y = m\,x + c$

$-1x + 2y - 2 = 1$

$-1x + 2y - 2 = 1$

$-1x + 2y = 1 + 2$

$-1x + 2y = 3$

$2y = 1x + 3$

$y = \frac{1}{2}\,x + \frac{3}{2}$

$y = 0.5\,x + 1.5$

$-1x + 2y - 3 = 0$

$-1x + 2y - 3 = 0$

$-1x + 2y = +3$

$2y = 1x + 3$

$y = \frac{1}{2}\,x + \frac{3}{2}$

$y = 0.5\,x + 1.5$

$-1x + 2y - 2 = -1$

$-1x + 2y = -1 + 2$

$-1x + 2y = 1$

$2y = 1x + 1$

$y = \frac{1}{2}x + \frac{1}{2}$

$y = 0.5\,x + 0.5$

$-1x + 2y\ -1 = 0$

$-1x + 2y = 1$

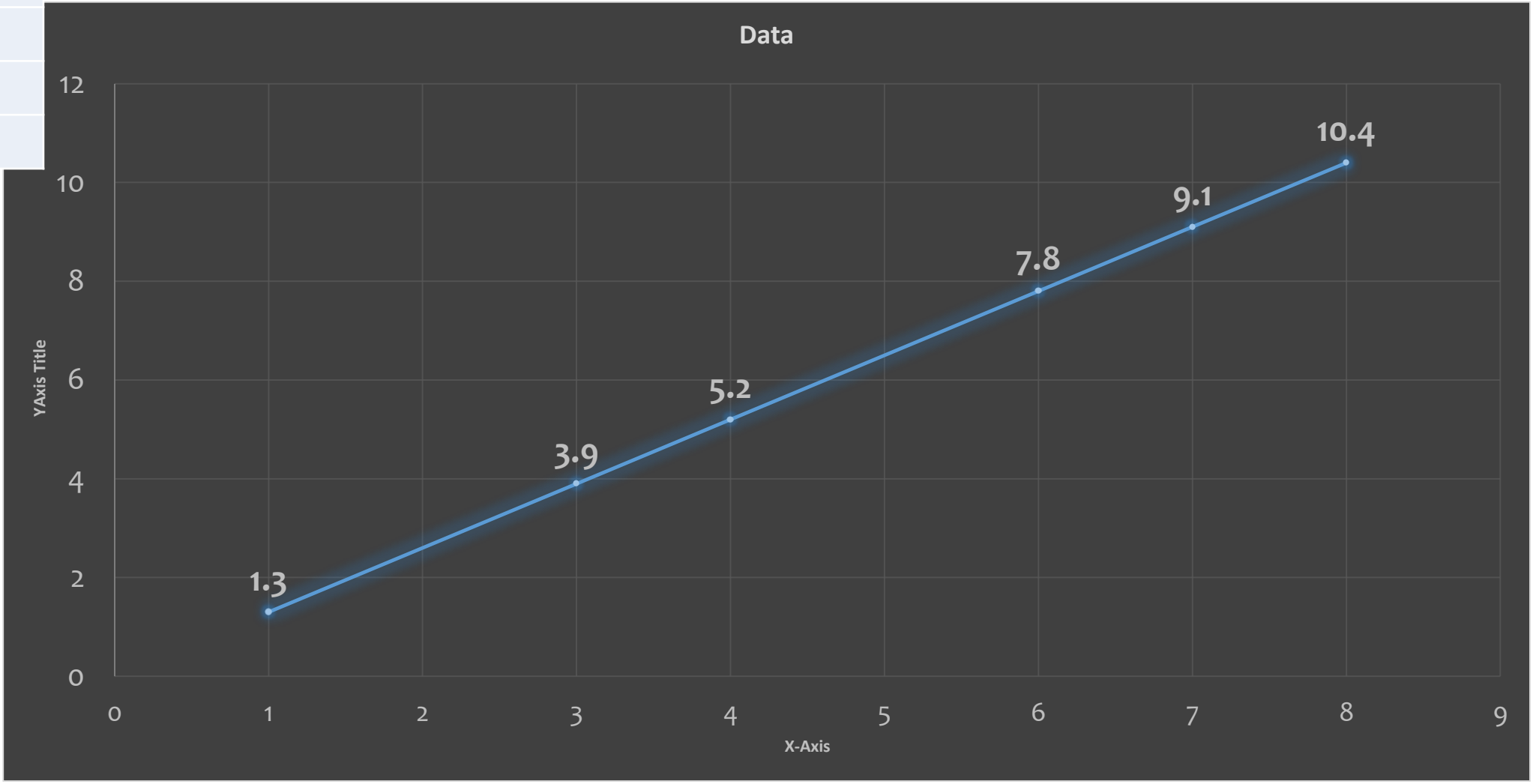
$2y = 1x + 1$

$2y = 1x + 1$

$y = \frac{1}{2}x + \frac{1}{2}$

$y = 0.5\,x + 0.5$

X	Y
X = 1	y = 1.3
X = 3	y = 3.9
X = 4	y = 5.2
X = 6	y = 7.8
X = 7	y = 9.1
X = 8	y = 10.4



Equation of Straight Line

$$ax + by + c = 0$$

y = m x + c

Equation of Straight Line

$ax + by + c = 0$

X=[1,3,4,6,7,8]

If

a = 1.3
b = -1
C = 0

x	(ax+by+c=0) (Substitute values)	Output
1	(1.3(1) - y + 0 = 0)	(y = 1.3)
3	(1.3(3) - y + 0 = 0)	(y = 3.9)
4	(1.3(4) - y + 0 = 0)	(y = 5.2)
6	(1.3(6) - y + 0 = 0)	(y = 7.8)
7	(1.3(7) - y + 0 = 0)	(y = 9.1)
8	(1.3(8) - y + 0 = 0)	(y = 10.4)

```
1.3(1)+(-1)*y + 0    = 0
1.3-y + 0            = 0
1.3-y                = 0
-y                   = 0-1.3
-y                   = -1.3
 y                   = 1.3
```